

Question	Marking Guidance	Mark	Comments
08.1	1. Unknown/new/different microorganisms/pathogens/microbes/bacteria (introduced); 2. (these bacteria) use food/space OR (these bacteria) produce toxins;	2 (2 x AO2)	1. Ignore chemical contaminant 2. Accept description of competition for other resources

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08.2	Correct answer, for 2 marks = 480 000;; Accept for 1 mark, 96 000 (correct number mm^{-3}) 120 (correct number in 0.000 25 mm^3 undiluted) OR any one of: 0.2 $\times 5 \frac{3}{15}$ (correct dilution factor) OR Evidence of dividing by 0.000 25	2 (2 x AO2)	$4.8 \times 10^5 = 2$ marks

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08.3	Correct answer for 3 marks, 3 bars given 1. Correct labelling of x axis; eg culture 1, culture 2 and culture 3 OR 1,2,3 and (bacterial) culture; 2. Height of culture 2 bar about <u>one fifth</u> height of culture 1 bar; 3. Height of culture 3 bar higher than culture 2 bar and lower than culture 1 bar; If 2 bars given OR if histogram given OR if graph given, accept for 2 marks, 1. Correct labelling of x axis; 2. Correct relative height of 2 bars/coordinates; eg height of culture 2 about <u>one fifth</u> height of culture 1 OR height culture 3 above height of culture 2 If no bar chart/histogram/graph given, accept for 1 mark, 1. Correct labelling of x axis;	3 (3 x AO2)	For labelling of x axis Accept 'no substance' for culture 1 Accept 'substance J' for culture 2 Accept 'substance K' for culture 3 Accept 'substance' for culture